

Biomaterial Habitats and Self-Replicating Systems Beyond Earth

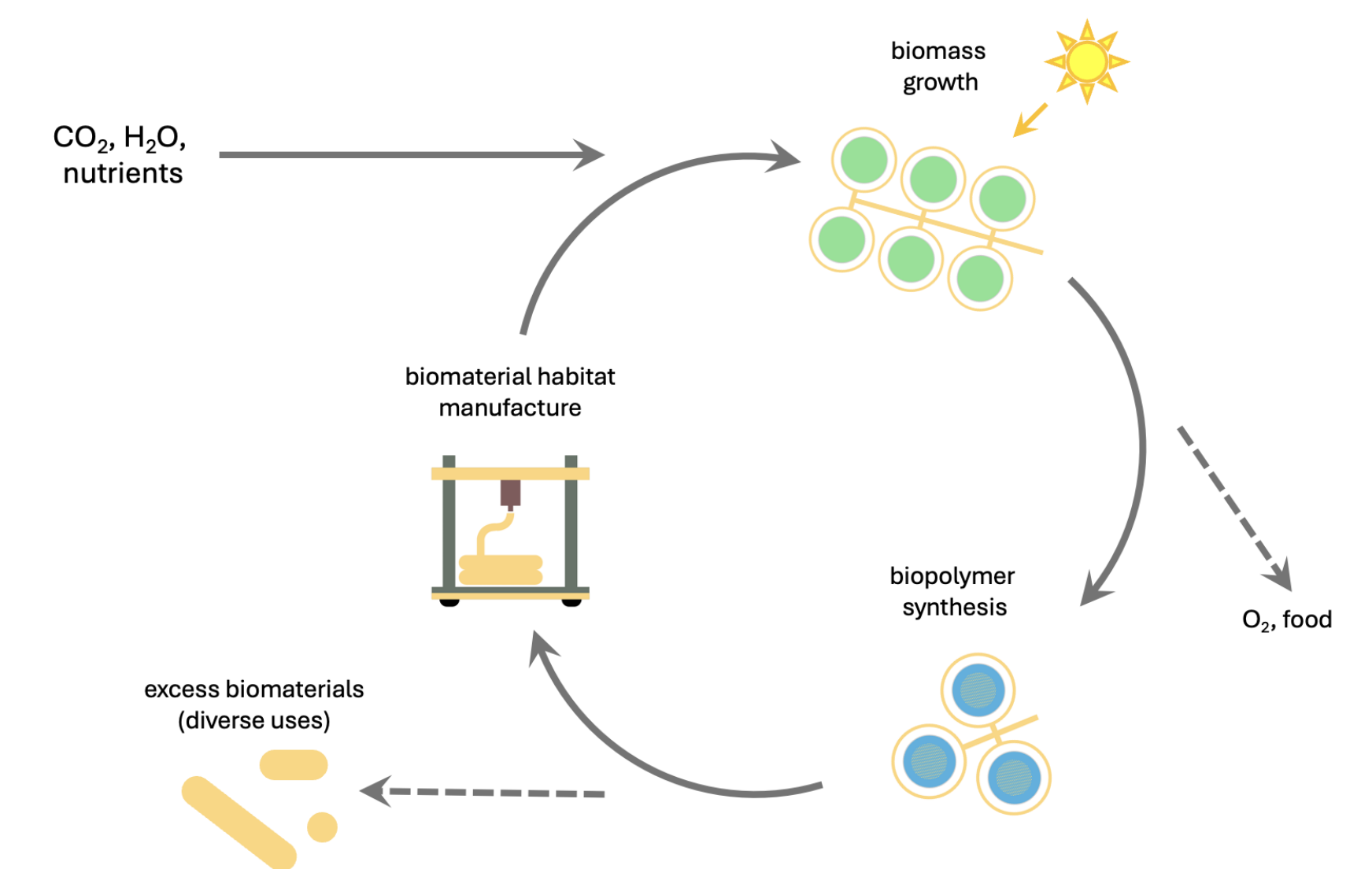
R. Wordsworth, R. Quayum, E. Kocharian, A. Pearson, X. Portillo, M. Yang, C. Cockell, G. Church & S. Nangle

Introduction

- Earth is the only known inhabited planet
- Standard definitions of habitability assume that life requires Earth-like conditions
- This is contradicted by human spaceflight: We are living organisms too
- How much technology / capability is needed in general for life to exist in non-terrestrial environments?

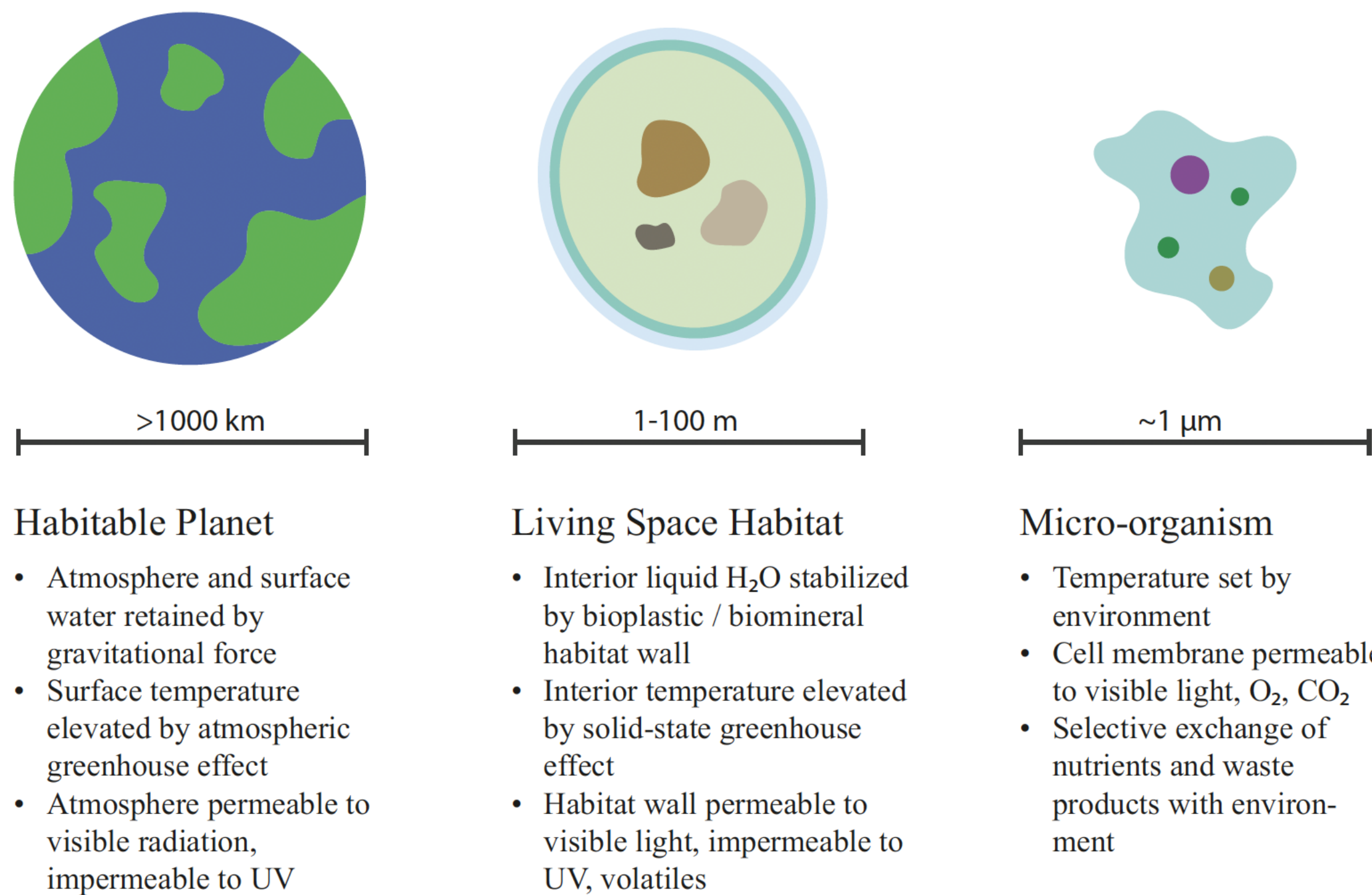
Implications for Extraterrestrial Life Support

- Idea allows controlled growth starting from small footprint: Photosynthesis → polysaccharides → more habitat material
- Opens a path towards scalable and *sustainable* human life support beyond Earth



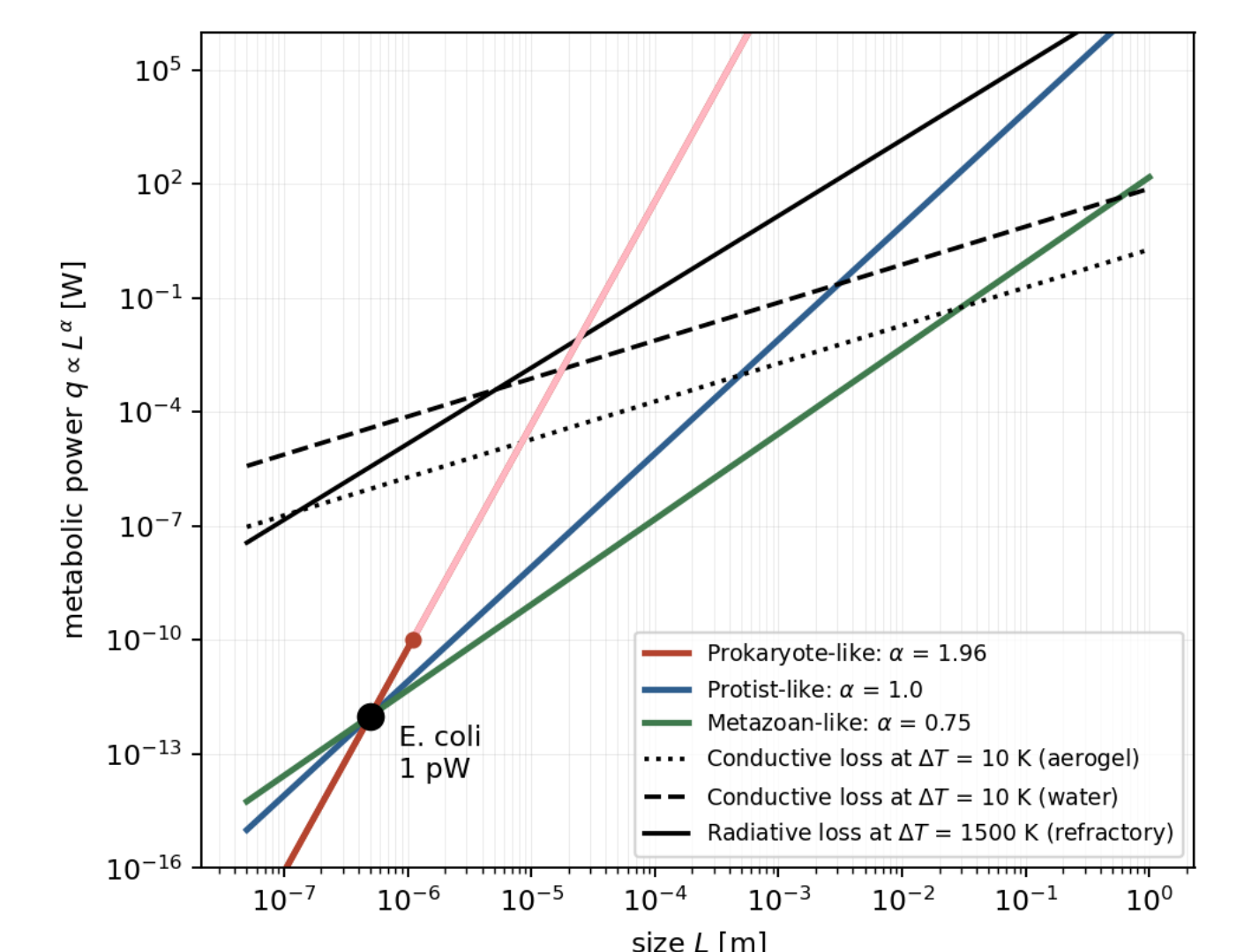
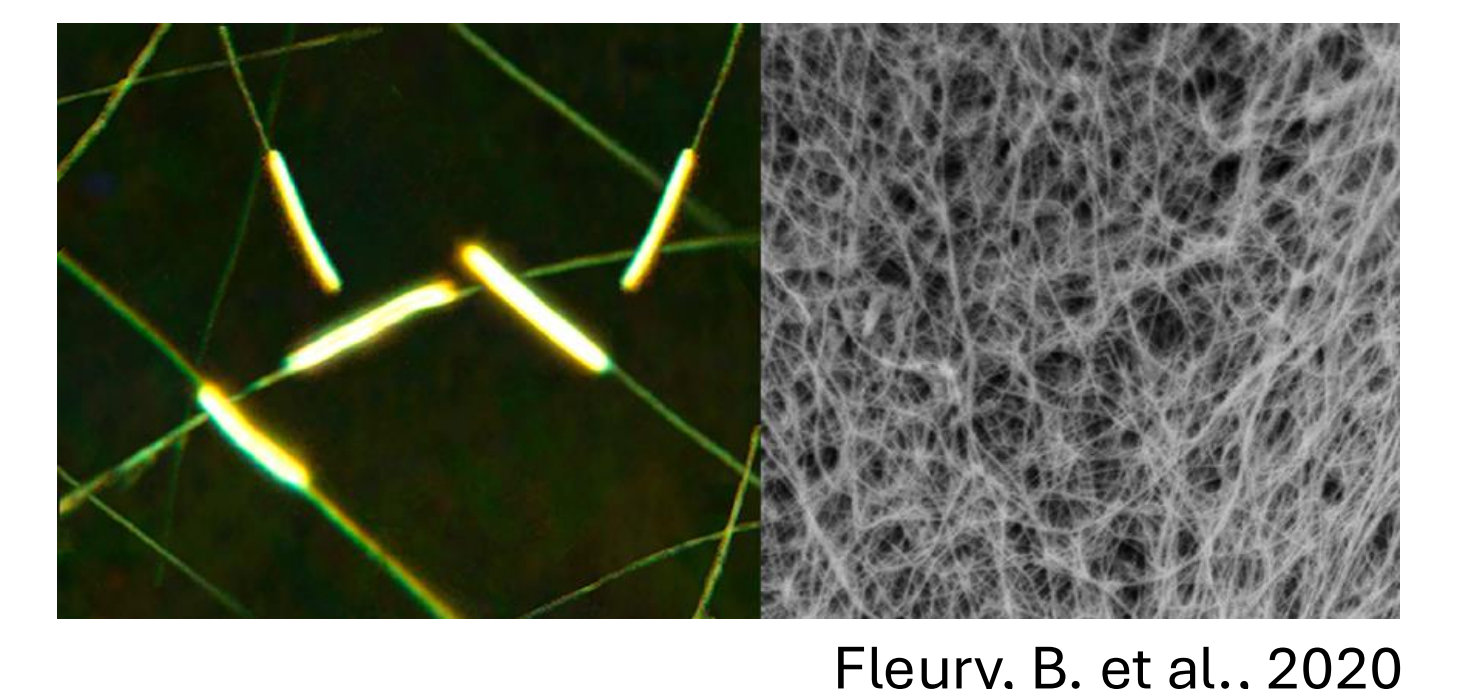
Thought Experiment

- Limit analysis to carbon-based life initially
- What would an organic structure capable of maintaining habitable conditions beyond Earth look like?



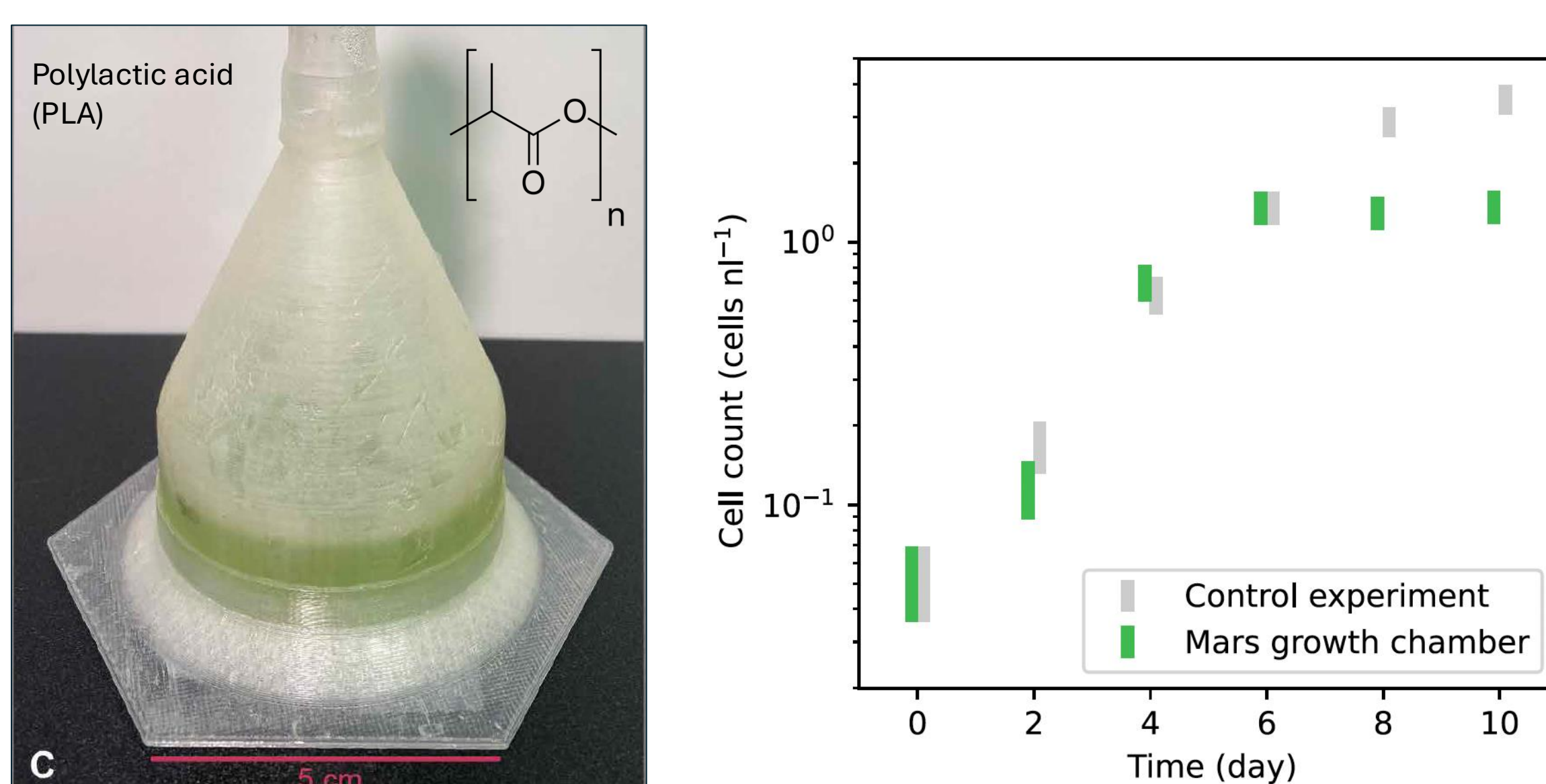
Temperature, Scale and Generalization

- Organic nanocellulose aerogels permit temperature regulation via solid-state greenhouse effect [3]
- Heat loss scales with size as $q \propto L^n$ where $1 \leq n \leq 2$
- Exerts strong control on minimum replicator scale in extreme environments
- Implications for more general systems (c.f. von Neumann 'clanking replicators')



Laboratory Experiment

- 3D-printed bioplastic habitats enable green algal growth under Mars-like pressure and atmospheric composition
- All habitat materials are biological in origin (polysaccharides, palmitic + abietic acids)



Conclusions

- Algal growth in hostile extraterrestrial environments possible in habitats made only of organic materials
- No physical or chemical limits on carbon-based life across a wide range of conditions: our current definitions of habitability are probably much too narrow
- Future work should systematically compare physics of different replicator classes in diverse environments